

DEXTER FERNANDES

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SUMMARY

Computer vision and perception engineer with 3+ years shipping real-time vision systems on edge hardware, plus an MSc in Robotics building **LiDAR** SLAM on a physical robot platform. Develops and tests image processing algorithms in **Python** (OpenCV) and **C++**, working with LiDAR and camera data for navigation and **localisation**. Background spans detector training and optimisation in PyTorch, point cloud processing, and integrating perception software with cross-functional hardware and software teams.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end **computer vision** lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in **Python** and **PyTorch**, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Designed adversarial-condition training experiments covering occlusion, lighting variation and motion artefacts, benchmarking robustness on F1 and failure rate to reach the 15% F1 gain in 3 months.
- Applied **image processing** and artefact detection with **OpenCV** and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Developed and maintained NVIDIA DeepStream and FFmpeg video pipelines for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics in desert conditions.
- Ran TIDE error analysis on COCO-format production data to split live detector failures by error type, directing dataset fixes and threshold tuning at the dominant one.
- Applied Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals in Jupyter to support model selection, threshold tuning and failure analysis.
- Led on-site validation on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous **C++17** perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames, cutting duplicate incident alerts from the moving-vehicle feeds.
- Implemented optical-flow based ego-motion estimation feeding CNN classifiers in PyTorch to infer vehicle state, supporting collision avoidance on moving vehicles.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Used confusion-matrix failure clustering and TIDE error analysis on COCO-format production data to diagnose recurring detector failures and prioritise dataset fixes across camera feeds.
- Worked with hardware, software and QA teams to integrate perception modules into deployed systems and shape rollout decisions.

PROJECTS

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in **ROS2** Jazzy and Gazebo Harmonic to evaluate **LiDAR**-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and **localisation** behaviour to support system selection.
- Implemented **point cloud** filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot_localization performing **sensor fusion** of wheel odometry and IMU.
- Developed ROS2 nodes, launch files and parameterised configs in **C++** and **Python**, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent camera feeds, with metadata extraction, anonymisation and re-streamed output.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Configured Ceres, g2o and GTSAM as solver backends and tuned ICP parameters across environments.
- Configured Nav2 for autonomous navigation and waypoint traversal in simulation over the generated map, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, SQL
Computer Vision & DL	object detection, image classification, semantic and instance segmentation, object tracking, optical flow, CNNs, YOLO (v5/v8/v10/v26), PyTorch, TensorFlow, Keras, OpenCV, scikit-image, NumPy
Robotics & SLAM	ROS2 (Robot Operating System 2), rclcpp, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, Nav2, LiDAR SLAM, pose graph optimisation, ICP (Iterative Closest Point), PCL (Point Cloud Library), Open3D, Ceres, g2o, GTSAM, robot_localization, EKF
Edge Inference	TensorRT, OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, NVIDIA Jetson AGX Xavier, Intel Movidius VPU
Video & Sensor Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-camera synchronisation, sensor calibration, ISP and camera tuning (RGB and IR)
Systems & Delivery	requirements-driven development, system integration across hardware and software, Agile and Scrum, stakeholder communication, cross-functional delivery, technical leadership and mentoring

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell — Service Champion / Team Trainer 09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)